

EXPERIMENT – 5a

UI using Axure RP

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AIM

To develop a mobile application prototype incorporating both familiar and unfamiliar navigation elements and evaluate usability among different user groups using Axure RP.

PROTOTYPE OVERVIEW

A mobile application prototype named ParkPilot was created using Axure RP. The prototype simulates a parking assistance system and incorporates both conventional and gesture-based navigation elements to analyze user interaction behavior. The design follows the Rapid Application Development (RAD model), enabling iterative design, testing, and refinement.

SCREEN-WISE DESCRIPTION

Intro / Onboarding Screen — “Get Started”

This is the initial screen displayed when the application launches. It introduces the app and prompts the user to proceed.

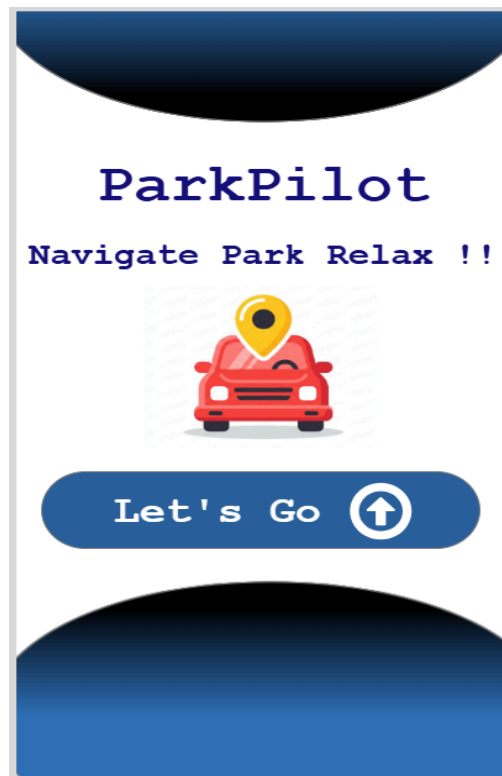
Familiar Navigation Elements

- Minimal onboarding layout similar to modern applications
- Clear instruction guiding progression
- Structured visual hierarchy

Unfamiliar Navigation Elements

- Swipe-up gesture used instead of a button
- No visible clickable control
- Requires interpretation of gesture-based interaction

This screen evaluates the user's ability to understand gesture-based navigation.



Authentication Screen — Login / Sign-Up

This screen allows users to either log in or create a new account.

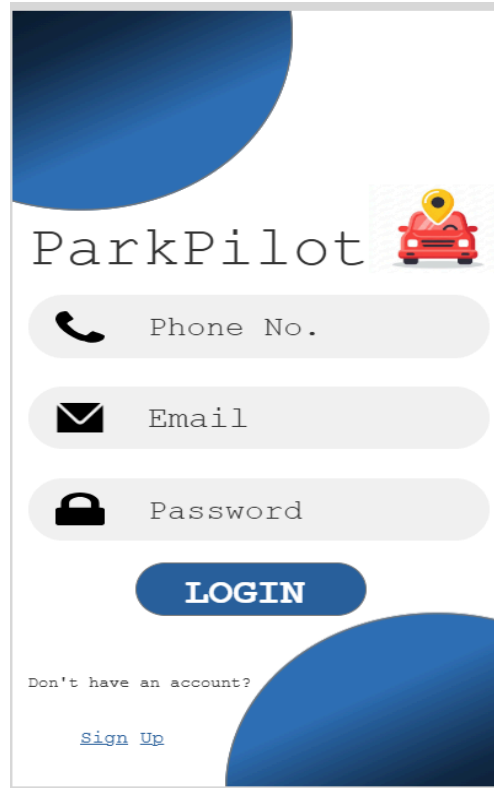
Familiar Navigation Elements

- Standard login and sign-up interface
- Clearly labeled buttons
- Tap-based navigation

Unfamiliar Navigation Elements

- First-time users must decide between login and registration
- Some users may hesitate without prior experience

This screen primarily relies on conventional UI patterns.



Account Creation Screen — “Let’s Begin”

This screen simulates account setup and progression.

Familiar Navigation Elements

- Sequential flow similar to typical onboarding processes
- Clear indication of next step

Unfamiliar Navigation Elements

- Swipe-right interaction replaces traditional “Next” button
- Gesture-based continuation may not be intuitive

This screen encourages exploratory interaction.



Home Dashboard Screen

This serves as the main interface providing access to core features.

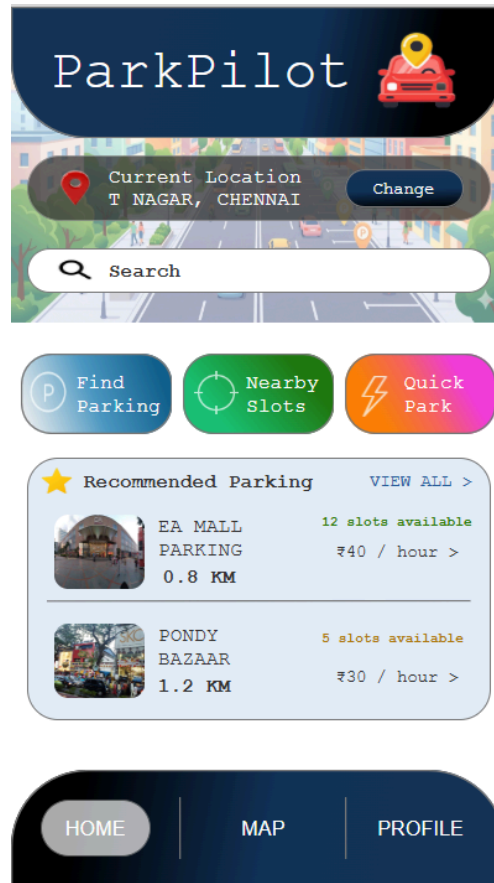
Familiar Navigation Elements

- Dashboard layout commonly used in mobile applications
- Clearly labeled navigation options (Map, Profile)
- Tap-based interactions

Unfamiliar Navigation Elements

- No guided tutorial for first-time users
- Users may explore before understanding functionality

This screen supports user orientation within the application.



Map Screen — Simulated Navigation View

This screen displays a visual representation of parking locations.

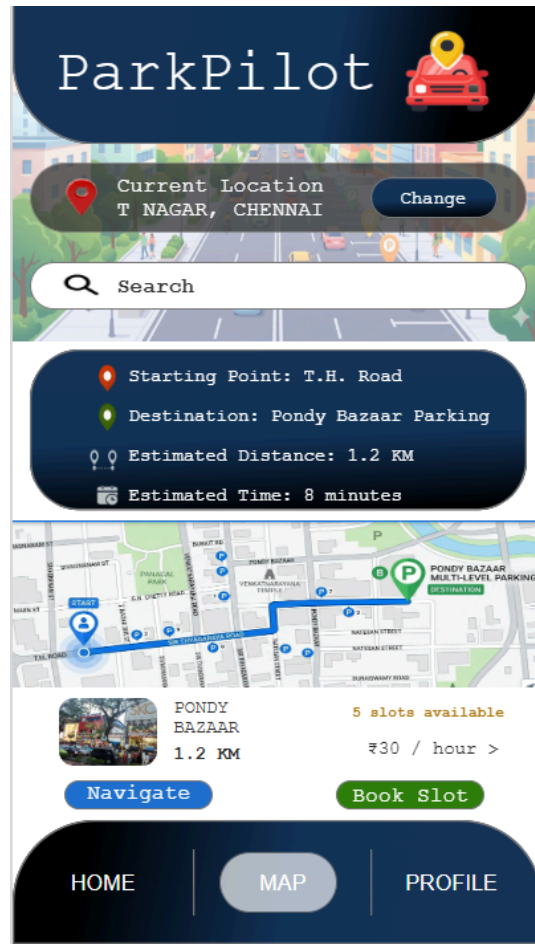
Familiar Navigation Elements

- Map-style UI similar to navigation applications
- Location markers and route visualization
- Spatial layout aiding understanding

Unfamiliar Navigation Elements

- Map is static and non-interactive
- Users may expect zooming or real-time updates
- Interpretation of symbols required

This screen evaluates spatial understanding and expectation alignment.



Profile Screen

Displays user information and account options.

Familiar Navigation Elements

- Standard profile layout
- Clearly structured information
- Logout option included

Unfamiliar Navigation Elements

- Users unfamiliar with navigation hierarchy may take time to locate features

This screen demonstrates hierarchical navigation structure.

Logout Function — Return to Login Screen

This function terminates the session and redirects the user.

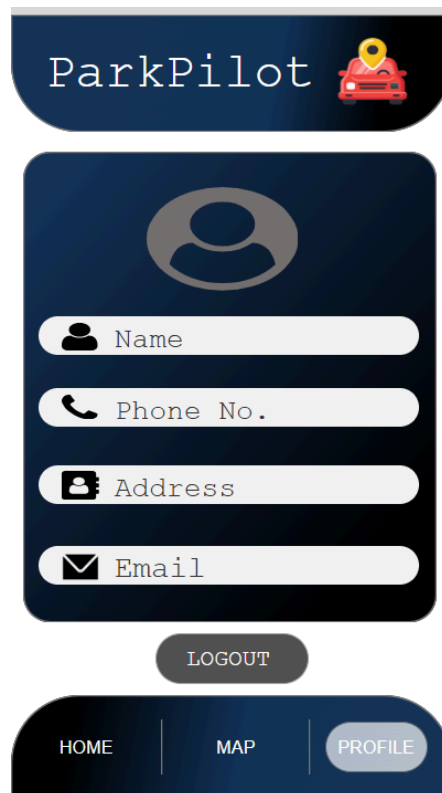
Familiar Navigation Elements

- Expected logout behavior
- Clear session termination

Unfamiliar Navigation Elements

- Immediate loss of session context may confuse new users

This screen reflects system state transition.



AXURE-SPECIFIC IMPLEMENTATION (RAD BASED)

Phase 1: Requirements Planning

- Identified features: parking search, booking, navigation, authentication
- Defined user actions: login, explore map, select slot, book parking
- Documented use cases for ParkPilot

Phase 2: User Design

- Created wireframes for all screens using Axure widgets

- Designed layouts for onboarding, login, dashboard, and map
- Added interactions using “OnClick” and gesture simulations
- Used Masters for reusable components (headers, navigation)
- Added annotations to describe UI elements and functionality

Phase 3: Construction

- Converted wireframes into interactive prototypes
- Implemented navigation flow:
 - Intro → Login → Home → Map → Slot → Profile
- Used dynamic panels for interactive behavior
- Tested prototype using preview mode
- Iteratively refined based on feedback

Phase 4: Cutover

- Finalized all screens and interactions
- Exported prototype as HTML / shared via Axure Cloud
- Prepared documentation for user guidance
- Conducted basic user orientation and testing

USABILITY EVALUATION CONSIDERATION

The prototype integrates both familiar and unfamiliar navigation elements to analyze user behavior. Familiar elements improve efficiency, while unfamiliar elements test adaptability and learning. Evaluation across different user groups helps understand the impact of prior experience on usability.

CONCLUSION

The ParkPilot prototype developed using Axure RP successfully demonstrates the application of the RAD model in UI design. By integrating familiar and unfamiliar navigation elements, the prototype highlights how different design approaches influence usability. The iterative development process ensures continuous improvement and effective user interaction without requiring full system implementation.